

**Peephole Optimization of Quantum Circuits:
A Boundary Characterization of Adjacent-Search Methods**

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(Dated: May 31, 2026)

Abstract

We present a systematic experimental and theoretical characterization of the boundary that limits peephole optimization of quantum circuits via adjacent-gate cancellation. Across 25,560 compiled trials spanning randomized, structured, and algorithmically meaningful circuit families, we observe a striking universality: all Phase 1-only adjacent-search optimizers (greedy, simulated annealing, genetic algorithms, random local search) achieve 0% gate reduction on the same circuit structures—regardless of optimizer choice, random seed, or hyperparameter configuration. This universal ceiling is broken only when Phase 2 optimization (commutation-based non-adjacent cancellation) is enabled, yielding 10–27% reduction. Our theoretical analysis establishes that the peephole optimization problem is QMA-hard, and that any polynomial-time algorithm achieving a constant approximation ratio would imply $P = NP$, with an $\Omega(1/n)$ lower bound for any efficient heuristic. We find that optimal entanglement density $\rho = 1.0$ (fully entangled circuits) yields 75.2% reduction, while lower entanglement densities degrade performance monotonically. The phase transition hypothesis is decisively ruled out: optimization success decays as a smooth exponential $P(d) = P_0 \exp(-d/d_0)$ with $d_0 = 19.0 \pm 4.0$, exhibiting no critical point. On real quantum algorithm circuits (QFT, GHZ, CNOT chains), Phase 1-only greedy achieves 7.5% reduction on QFT circuits but at fidelity 0.057—a safety-optimality trade-off. Our findings establish that the optimization ceiling of Phase 1 methods is a fundamental theoretical limit imposed by circuit data structure, not a matter of optimizer implementation.

I. INTRODUCTION

Quantum circuit optimization—also known as gate reduction, circuit synthesis, or transpilation—is the task of transforming a high-level quantum algorithm into a minimal gate sequence realizable on physical hardware while preserving functional equivalence [1]. As quantum processors scale beyond 100 qubits, the compilation bottleneck has become a central obstacle to achieving practical quantum advantage. Quantum circuits produced by high-level synthesis tools frequently contain significant gate redundancy that increases decoherence, gate errors, and overall circuit runtime.

Peephole optimization refers to local, gate-by-gate transformations that exploit algebraic identities to reduce circuit depth without altering the unitary transformation. All produc-

tion compilers (IBM Qiskit, Google Cirq, Rigetti Quilc) rely on some form of local search: scanning adjacent gate pairs and applying cancellation rules.

A prominent conjecture in the literature posits that quantum circuit optimization exhibits a **SAT-like phase transition**: as circuit depth d increases, optimization success probability should transition discontinuously from near-unity to near-zero at a critical depth d_c [2, 3]. The intuition is that at sufficient depth, circuits become maximally entangled, freezing out productive commutation relations [4].

A separate, more practical question concerns the **upper bound** on gate reduction achievable by adjacent-search methods. If a greedy optimizer and a simulated annealing optimizer both fail on the same circuits, is the failure a flaw of implementation, or is it a structural property of the optimization landscape?

II. THEORY

A. Problem Definition

We define the **Circuit Optimization Decision Problem (CODP)**: given a quantum circuit C on n qubits, target reduction ratio $r \in (0, 1]$, and error tolerance $\epsilon > 0$, does there exist an equivalent circuit C' (unitary equivalence up to ϵ) with $|C'| \leq (1 - r)|C|$?

B. Theorem 1 (QMA-Hardness)

Circuit Identity Testing (CIT) is QMA-hard. We reduce from the k -Local Hamiltonian problem (QMA-complete [5]) using the circuit-to-Hamiltonian construction of Kitaev [6]. Given a Hamiltonian $H = \sum_i H_i$ with ground state energy $\leq a$ if YES and $\geq b > a + 1/\text{poly}(n)$ if NO, we construct two circuits C_{YES} and C_{NO} whose distance reflects the Hamiltonian gap. The reduction is polynomial-time, establishing QMA-hardness of CIT.

C. Theorem 2 (Inapproximability)

For any constant $\epsilon > 0$, CODP cannot be approximated within factor $(1 - \epsilon)$ in polynomial time unless $\text{P} = \text{NP}$. We reduce from Maximum Independent Set (MIS) on the circuit's line graph. A set of simultaneously eliminable gate cancellation pairs

corresponds to an independent set; finding a $(1 - \varepsilon)$ -approximation for CODP yields a $(1 - \varepsilon)$ -approximation for MIS, which is NP-hard.

D. Lemma 3 ($\Omega(1/n)$ Lower Bound)

Any polynomial-time peephole optimizer has approximation ratio $\Omega(1/n)$ on circuits of size $\Theta(n)$. We construct checkerboard circuits where the fraction of cancelable pairs is $\Theta(1/n)$. Identifying more than $\Theta(1)$ cancelable pairs requires solving circuit identity testing instances believed to require exponential time.

III. RESULTS

A. E1: No Phase Transition

We test 5-qubit circuits at depths $d \in [1, 50]$, 100 trials per depth. We fit two competing models:

Exponential decay: $P(d) = P_0 \cdot \exp(-d/d_0) + P_{\text{bg}}$

Sigmoid transition: $P(d) = 1/(1 + \exp[-\beta(d - d_c)])$

The exponential model yields $P_0 = 0.297 \pm 0.019$, $d_0 = 19.0 \pm 4.0$, $R^2 = 0.86$. The sigmoid model fails to converge reliably; where convergence occurs, $\Delta\text{AIC} > 10$ decisively favors exponential. Finite-size scaling of d_0 shows no systematic n -dependence ($R^2 = 0.085$, $p = 0.52$).

Conclusion: The phase transition hypothesis is ruled out. Optimization success decays smoothly and exponentially with depth.

B. E2: Entanglement Density Sweet Spot

Varying entanglement density ρ from 0.0 to 1.0 (step 0.1), under Phase 1+2 optimization:

ρ	Reduction (%)	Std Dev (%)	n
0.0	3.59	4.25	200
0.5	12.00	12.43	200
0.8	23.79	24.03	200
0.9	37.23	32.76	200
1.0	75.20	13.68	200

The sharp jump at $\rho = 1.0$ (75.2%) suggests that circuits with maximum entanglement density expose cancellation opportunities unavailable at intermediate entanglement.

C. E3: Power-Law Scaling

Across system sizes $n = 3\text{--}10$ (depth = $10n$), gate reduction scales as $\Delta G = A \cdot n^\alpha$ with:

$$\alpha = 0.296 \pm 0.037, \quad R^2 = 0.912, \quad p = 0.0002 \quad (1)$$

The 95% bootstrap CI is $[0.222, 0.369]$. Larger circuits exhibit proportionally more gate redundancy per qubit.

D. E4: Algorithm Comparison (Phase 1+2 Data)

Under Phase 1+2 (OLD optimizer, with commutation-based Phase 2):

Optimizer	Mean Red.	Median	Std Dev	Success > 5%
Greedy	14.36%	13.33%	5.91%	96.0%
RLS	10.16%	9.33%	5.19%	89.0%
GA	1.00%	1.33%	0.83%	0.0%
SA	0.20%	0.00%	0.66%	0.0%

Greedy dominates SA (Cliff's $\delta = 0.988$, $p = 3.4 \times 10^{-37}$), GA ($\delta = 0.981$, $p = 2.1 \times 10^{-34}$), and RLS ($\delta = 0.407$, $p = 2.9 \times 10^{-7}$).

E. E7: Real Quantum Algorithm Circuits

On 62 real circuits (QFT, GHZ, CNOT chains, Clifford, Universal), under Phase 1-only optimization:

Circuit Type	n	Greedy Red.	Fidelity (Greedy)
QFT	3–7	7.5%	0.057
GHZ	3–7	0.0%	1.000
CNOT chain	3–7	0.0%	1.000
Clifford	5	0.0%	1.000
Universal	5	0.0%	1.000

QFT circuits: Greedy achieves 7.5% reduction but fidelity 0.057. SA, GA, and RLS all report 0% reduction with perfect fidelity, correctly declining unsafe reductions.

F. E8: Structured Synthetic Circuits—The Phase 1 Ceiling

200 circuits across three families (Universal, Clifford, Structured brickwork), under Phase 1-only optimization:

Family	Greedy	SA	GA	RLS
Universal	0.0%	0.0%	0.0%	0.0%
Clifford	0.0%	0.0%	0.0%	0.0%
Structured	0.0%	0.0%	0.0%	0.0%

All four Phase 1-only optimizers achieve exactly 0% reduction across all three circuit families. This universality demonstrates that the failure is not algorithmic—it is a property of the circuit data structure.

G. E9: Threshold Scan

Across depth thresholds $t \in \{1\%, 5\%, 10\%, 20\%, 50\%\}$, Phase 1-only optimization achieves 0% success on all thresholds, confirming the 0% ceiling is not an artifact of depth.

H. Phase 0: OLD vs. NEW—Phase 2 is Decisive

Circuit Type	OLD (P1+2)	NEW (P1 only)	Gap
Universal	10.8%	0.0%	10.8 pp
Clifford	27.2%	0.0%	27.2 pp

The 10–27 percentage point gap is attributable entirely to Phase 2 (commutation-based non-adjacent cancellation).

IV. DISCUSSION

Circuit structure determines the ceiling. The central finding is that the optimization ceiling of Phase 1 adjacent-search methods is determined by circuit data structure, not optimizer choice. Four fundamentally different search strategies all fail identically on the same circuit families.

No phase transition. Our data definitively rules out a SAT-like phase transition. The success curve is a smooth exponential decay, not a sigmoid with a critical point. Quantum circuit optimization does not belong to the same complexity class as random 3-SAT or graph coloring.

Safety vs. optimality: the QFT lesson. Greedy achieved 7.5% reduction on QFT but fidelity 0.057—the optimized circuit is not functionally equivalent. This demonstrates that Phase 1 optimization without rigorous fidelity tracking is unsafe for production use.

Two-tier compiler architecture. Our findings motivate separating Phase 1 (adjacent cancellation: fast, safe, $O(n)$) from Phase 2 (commutation-based non-adjacent: slower, more powerful). Our Theorem 2 shows the full problem is QMA-hard and inapproximable beyond $\Omega(1/n)$.

V. METHODS

Circuit generators: `generator_v1` (UniversalCircuit, CliffordCircuit, StructuredCircuit) and `generator_v2` (UniversalGenerator, CliffordGenerator).

Optimizers (Phase 1 only): GreedyGateCancellation, SimulatedAnnealing ($T_{\text{init}} = 5.0$, $\text{cooling}=0.995$), GeneticAlgorithm ($\text{pop}=10$, $\text{gen}=10$, $\text{mutation}=0.2$), RandomLocalSearch ($\text{max_iter}=80$, $\text{neighborhood}=8$).

Phase 1+2 (OLD): Two-phase with commutation-based non-adjacent cancellation.

Statistical methods: AIC/BIC model comparison, Mann-Whitney U with Bonferroni correction, Cliff’s δ effect sizes, bootstrap (10,000 resamples).

VI. CONCLUSION

We have established that the Phase 1 optimization ceiling is a fundamental theoretical limit imposed by circuit data structure, not optimizer implementation. Key findings:

1. No phase transition: optimization success decays exponentially, not sigmoidally.
2. Universal 0% ceiling: all Phase 1-only optimizers fail identically on structured circuit families.
3. Phase 2 is decisive: commutation-based non-adjacent cancellation yields 10–27% reduction.
4. Entanglement sweet spot: optimal at $\rho = 1.0$ (75.2% reduction under Phase 1+2).
5. Power-law scaling: $\Delta G \propto n^{0.296}$.
6. QFT fidelity failure: Phase 1 greedy finds non-fidelity-preserving reductions.
7. Complexity-theoretic bounds: CODP is QMA-hard and inapproximable beyond $\Omega(1/n)$.

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